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Enhancing CO_2 Sequestration in Deep and Highly Saline Aquifers Using Chelating Agents

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Abstract

Deep saline aquifers are a critical target for large-scale CO_2 sequestration due to their widespread distribution, huge pore volume, and favorable geochemical properties. However, one of the key challenges in these aquifers is the lower CO_2 solubility and storage capacity caused by cations such as calcium and magnesium, which contribute to salt precipitation and pore blockage during CO_2 injection. This study investigates the use of chelating agents such as ethylenediaminetetraacetic acid (EDTA) and N, N-dicarboxymethyl glutamic acid (GLDA), as reservoir conditioners that selectively bind these cations, thereby preventing salt precipitation and enhancing CO_2 solubility in brine.

The research employs a combination of solubility tests and core flooding experiments to simulate aquifer conditions. Solubility tests were conducted under high-pressure, high-temperature conditions representative of deep aquifer environments in a pressure-volume- temperature (PVT) system, using synthetic brine/seawater formulations treated with varying concentrations of EDTA and GLDA. These tests assess the agents' effectiveness in sequestering targeted cations and increasing CO_2 dissolution. Complementary analytical techniques, such as inductively coupled plasma (ICP) spectroscopy, assisted in quantifying dissolved cation concentrations. The results showed that pH and salinity play critical roles in the solubilization of CO_2 in brine. Both GLDA and EDTA performed on par; however, GLDA was able to chelate slightly more ions and promote solubility. Furthermore, core flooding experiments will be carried out using carbonate-rich rock samples to evaluate the impact of chelating agents on CO_2 injectivity, mineral-fluid interactions, and pore structure evolution. These dynamic tests assess the performance of these chelating agents towards improving aquifer storage capacity in flow conditions. X-ray computed tomography (CT) will be used to visualize changes in pore structure and mineral distribution, sonic velocity measurements for tracking changes in the mechanical properties of rock samples, and low-field nuclear magnetic resonance (NMR) for evaluating pore size profile evolution.

This novel approach acknowledges the environmental concerns of chelating agent degradation in aquifer conditions, as well as the cost- effectiveness of the squeeze treatment to enhance the CO_2 storage through chelation in a highly saline formation, ensuring a sustainable and promising technique for global CO_2 reduction.

Keywords: CO_2 sequestration, chelating agents, saline aquifers, solubility, salinity, ion concentration

Introduction

CO_2 storage in deep saline aquifers is a technique that has gained worldwide attention over the last few decades. Herein, atmospheric CO_2 is injected deep into saline geological formations, in which brine and formation minerals help solubilize and subsequently mineralize carbon dioxide, successfully immobilizing it underground (Bashir et al., 2024). They are preferred over other subsurface options due to the abundance of water or brine and low-permeability caprock that traps CO_2 within the aquifer (Mim et al., 2023). CO_2, once injected in these aquifers, reacts with the formation brine/fluid and forms a dense phase, which allows it to settle in the aquifer and further mineralize upon reactions with the surrounding formation ions (Luo et al., 2022). Moreover, the global storage capacity of deep saline aquifers is known to be more than 1000 gigatons, which is enormous compared to other geological storage media (Benson and Cole, 2008). Suitable depths for these formations are 800-3000 m, where the supercritical state of CO_2 is met at 31.1°C and 1071 psi, making the gas attain an intermediate phase between gas and liquid that ensures efficient trapping without leakage (Celia et al., 2015). It is also necessary to characterize these storage sites to ensure proper entrapment and prevent groundwater contamination (Metz et al., 2005).

There are various mechanisms through which CO_2 is stored in a saline aquifer, among which solubilization is a primary process. After undergoing residual trapping, CO_2 mixes with the formation brine to form a dense phase and settles at the aquifer basin, thereafter which it can be permanently immobilized via mineralization (Bashir et al., 2024). However, the solubility of CO_2 relies on various factors such as the brine's salinity, pH, temperature and pressure of the formation (Edem et al., 2020; Khanal et al., 2024; Rosenbauer et al., 2005). Out of these properties, the brine salinity dictates how fast CO_2 is dissolved and accommodated by the fluid in a specific region (Mouallem et al., 2024). The higher the salinity, the lower the dissolution of CO_2 will be as salt ions compete with CO_2 at the molecular level for their place in brine (Pruess and Muller, 2009). Moreover, salinity cannot be altered; rather, we can inject and expose the brine to certain treatments that reduce the effect of salinity that hinders CO_2 solubility.

Freshwater injection is a common and economic practice to dilute the brine and reduce the local salinity, while maintaining formation permeability (Darkwah-Owusu et al., 2024; Pruess and Muller, 2009). Though freshwater's reactions with rock minerals would trigger injectivity impairment causing more bad than good. Nanoparticles (NPs) have gained attention as a powerful option in CO_2 storage as they can alter the fluid-fluid and rock-fluid interactions at the pore level. Scientists have explored the use of silica, lead, and uranium dioxide (UO2) NPs in forming stable CO_2 foams under tough reservoir conditions and their ability to elevate the density of CO_2-brine phase, accelerating convective mixing (Alzobaidi et al., 2017; Javadpour and Nicot, 2011). Moreover, their manufacturing costs and long-term stability in subsurface conditions proved to be a few challenges in their continued use in CO_2 storage. Surfactants used to create CO_2 foams have long been used for enhanced oil recovery to control CO_2 mobility and improve oil/gas sweep efficiency in production (Massarweh and Abushaikha, 2020). In CO_2 storage, surfactants such as Tergitol 15-S-9 and Igepal CO-720 proved beneficial in producing strong and stable CO_2 foams over many injected pore volumes. However, foam decay due to adsorption and stripping on the CO_2 phase must be considered before the selection of surfactants (Føyen et al., 2020).

Chelating agents such as N,N-dicarboxymethyl glutamic acid (GLDA) and ethylenediamine tetraacetic acid (EDTA) have emerged as a medium to extract magnesium or calcium ions from silicates to improve mineral carbonation (Zhao et al., 2013). They have already performed effectively in various applications in the oil and gas industry, ranging from enhanced oil recovery to well productivity and scale removal (Hassan et al., 2020). Being well-known compounds in the field of coordination chemistry, their ability to bind cations has made them suitable candidates for metal extraction, such as magnesium from serpentine

(Wang et al., 2021). Moreover, the efficiency of how well a ligand or chelating agent can bind a metal cation depends on the stability constant of the metal complex. In short, the binding strength of a chelating agent is directly proportional to its stability constant (Martell and Hancock, 1996). The valency of the metal cation is also responsible for deciding how strongly the ligand can bind to it; examples include Ca^{2+}, Mg^{2+}, and those with charges 2 or above will have a candidacy for efficient binding (Almubarak et al., 2017). However, the widespread use of these chelating agents in improving CO_2 solubility in saline aquifers is yet to be employed, mainly due to the degradation concern in subsurface conditions over long periods (Wang et al., 2024). Furthermore, the cost of GLDA (~1.20$/kg) is higher than that of conventional acids, so these chemicals should be used in considerate amounts (Wang et al., 2022). Sodium chloride (NaCl), being a prominent salt in a saline environment, also poses a challenge for chelating agents since Na^+ is a single-charge cation. However, this can be resolved using a calcium-based EDTA to chelate sodium ions (Fredd and Fogler, 1998). The application of EDTA to enhance CO_2 mineralization in sour brines was studied by Mohammed et al. (2025) while considering the impact of hydrogen sulfide (H_2S) on the same process. It was implied that ETDA improved CO_2 solubility and mineralization via cation dehydration and reducing their interference in saline brines. However, an excess of H_2S degraded EDTA over time, limiting its activity in sour conditions (Mohammed et al., 2025). Given in Table 1 are the stability constants of some cations and EDTA. In Fig. 1, the speciation or fractional composition of EDTA with pH is given. As pH increases, EDTA forms more stable complexes and the stability increases from left to right with more deprotonated species, however, maintaining a pH < 9 is important to chelate divalent cations (Fredd and Fogler, 1998b).

This paper examines the use of chelating agents in a seawater of typical composition and compares it to a calcium chloride (CaCl2) brine of extremely high salinity (200,000 ppm), while evaluating the increase in solubility offered by the chelation process. The concentrations of chelating agents in brine were chosen to comply with the cost of these chemicals, as well as to address their degradation process in the aquifer.

Table 1—Stability constants of some cations with EDTA (Smith & Martell, 1987)

Cation	Stability constant (log K)
$Fe^{3}+$	25.1
$Cu^{2}+$	18.8
$Pb^{2}+$	18
$Zn^{2}+$	16.5
$Ni^{2}+$	18.6
$Co^{2}+$	16.3
$Ca^{2}+$	10.7
$Mg^{2}+$	8.7
Na+	1.67
K+	1.33

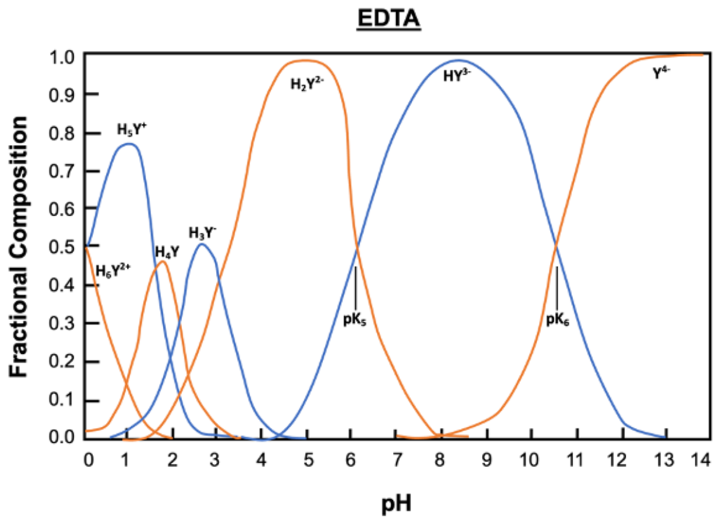

Figure 1—Fractional composition of EDTA with pH (Fredd and Fogler, 1998b)

Materials and Methods

Materials

The brine used was synthetic seawater (SW) with total dissolved solid (TDS) content of 57.28 g/l, of which the composition is listed in Table 2. EDTA of 99% assay was obtained from Sigma-Aldrich™, and GLDA of Dissolvine® StimWell HTF with 85% purity was used. The salts for brine preparation were obtained from various companies, while deionized (DI) water was produced using Millipore

Table 2—Synthetic seawater composition

Ions/Species	Concentration (g/l)
Chloride (Cl^-)	31.81
Sodium (Na+)	18.04
Sulfate ($SO4^{2-}$)	4.45
Magnesium (Mg^{2}+)	2.16
Calcium (Ca^{2}+)	0.65
Bicarbonate ($HCO3^-$)	0.17
Total Dissolved Solids (TDS)	57.28

Milli-Q water system. The pH of the EDTA solution was adjusted using sodium hydroxide (NaOH). An IKA mixer was used to prepare the solutions, while magnetic stirring plates from VELP were used to make SW. A densitometer from Anton Paar and a pH meter from Mettler Toledo were used for liquid examination. ICP-OES from Perkin Elmer was used for ion concentration measurement. A pressure- volume-temperature (PVT) system from Sanchez Technologies was used to examine the fluid-fluid interaction, whose schematic is given in Fig. 2. It has a 300ml cell that can be used as per set parameters for CO_2 and brine contact.

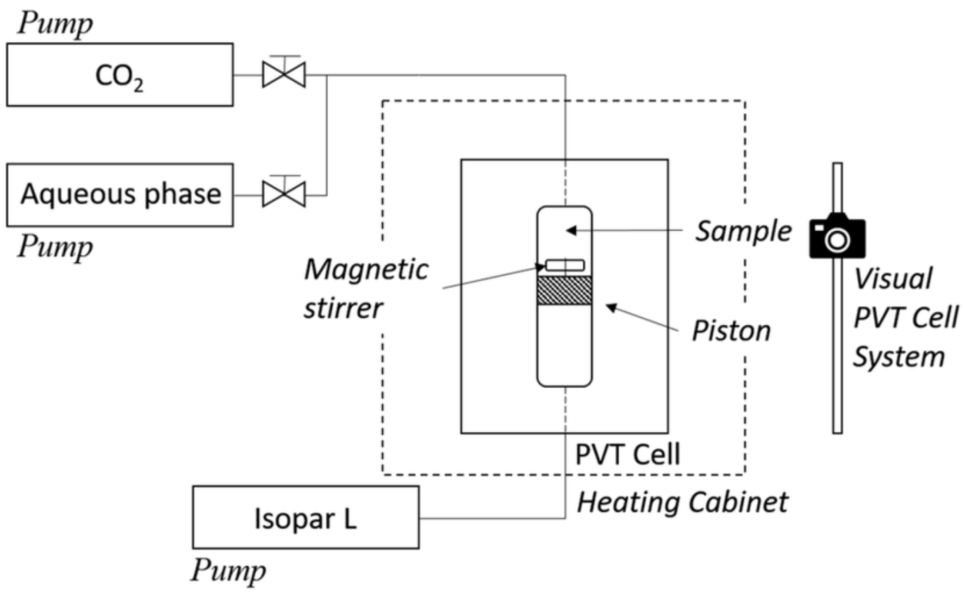

Figure 2—A schematic showing the different parts of a PVT cell (Barrabino et al., 2018)

Methods

The primary mechanism of the CO_2 injection with chelating agents is shown in Fig. 3, showing their effect in improving the CO_2 solubility near the injection well. In Fig. 3A, the CO_2 is injected solely without any additives, such that it forms a separate phase of gas in brine near the injection well. Meanwhile, in Fig. 3B, CO_2 is co-injected with chelating agents, which enhances its solubility in brine and transitions smoothly into the brine phase.

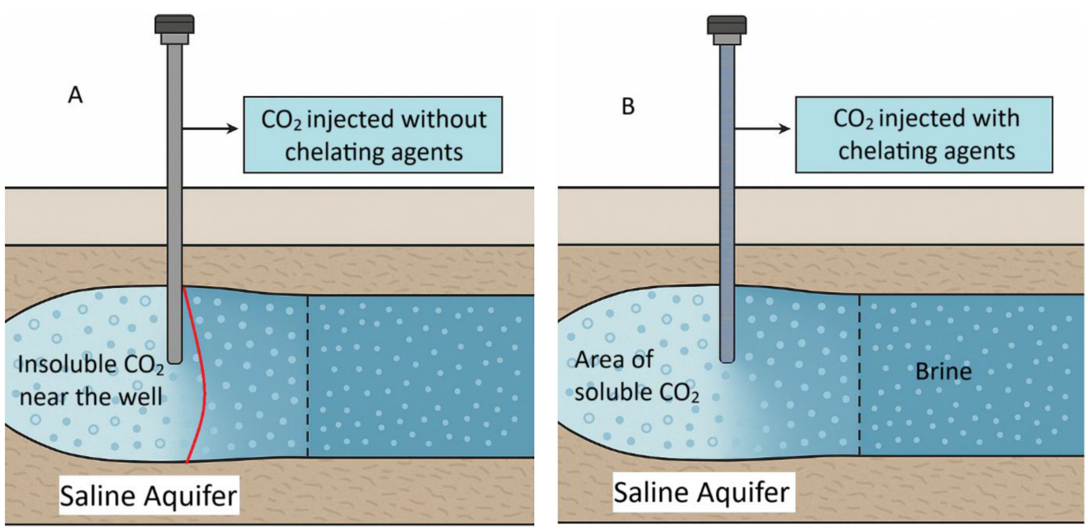

Figure 3—Solubility of CO_2 in brine injected A) without chelating agents and B) with chelating agents

The number of moles of CO_2 dissolved in the brine was calculated using the ideal gas law, given in Equation 1. It is used because the pressure-volume-temperature (PVT) system contains two fluids with only CO_2 in the gas phase.

$$PV = nRT \tag{1}$$

where, P is the pressure of the system in atm, V is the volume of CO_2 dissolved in brine in litres, n is the number of moles of CO_2 dissolved in brine, R is the universal gas constant, and T is the temperature in K.

The solubility of CO_2 was determined by using the molarity formula in Equation 2, which gives the number of moles per litre of brine.

$$Molarity\left(\frac{mol}{L}\right)=\frac{n}{Initial\ volume\ of\ brine} \tag{2}$$

This value can be obtained in mol/kg by dividing the molality by the brine's density.

Preliminary case. A 20% $CaCl_2$ brine solution was prepared by weighing the required amount of CaCl2 salt dissolved in DI water. This concentrated brine was used as the base fluid for all experimental samples. In parallel, two chelating agent solutions of EDTA and GLDA, each of 5% concentration, were prepared by dissolving a precise amount of each pure chemical in DI water, followed by thorough mixing to ensure homogeneity. Since both EDTA and GLDA are pH-sensitive and function optimally under alkaline conditions, the pH of each solution was carefully adjusted using NaOH to reach the desired basic range suitable for effective chelation of metal ions such as calcium.

Three different brine formulations were prepared for comparative analysis:

1. A control sample of pure 20% CaCl2 solution with no chelating agent,
2. A second sample with 20% CaCl2 with 5% EDTA,
3. A third sample with 20% CaCl2 with 5% GLDA.

For each sample, key physical properties, including pH and density (listed in Table 3), were measured before CO_2 injection for a baseline comparison. These prepared brine samples were loaded into individual accumulator cells designed to sustain high pressure and temperature, simulating subsurface conditions. The cells were sealed, and the remaining volume (headspace) was injected with CO_2 gas at a pressure of 1500 psi, ensuring direct contact with the brine. The cells were then placed in an oven and heated at a constant temperature of 50°C for 24 hours. This was done to simulate thermal conditions in an aquifer and promote CO_2 dissolution. During this period, the pressure was monitored using a pressure gauge for any fluctuations. After 24 hours, the cells were carefully depressurized and the effluent brine from each was collected. The volumes of brines were measured, and an increase in the liquid phase was attributed to dissolved CO_2. This metric was a direct measurement of the increase in CO_2 solubility due to chelating agents.

Table 3—Physical properties of brine samples (Preliminary case)

Sample	pH	Density (g/cc)
Control sample (20% CaCk)	9.56	1.13
20% CaCl2 + 5% EDTA	7.59	1.129
20% CaCl2 + 5% GLDA	3.25	1.13

To evaluate the chelation efficiency of EDTA and GLDA, a chemical analysis of the brine samples was conducted using inductively coupled plasma optical emission spectroscopy (ICP-OES). Calcium ion concentrations were measured before and after CO_2 interaction. A reduction in free calcium ions after CO_2 exposure indicated successful chelation due to reduced ion interactions and potential changes in brine chemistry.

The quantification of free ions (those are responsible for salt formation/precipitation) was done by ICP-OES using sodium carbonate (Na_2CO_3) as a buffering and precipitating agent. Initially, the total ion concentration that includes both free and chelated ions was determined using ICP. This was followed by adding a known amount of Na_2CO_3 to the same sample for 1:1 chelation and mixing, then put to rest for calcium carbonate ($CaCO_3$) to precipitate, formed by the reaction of free calcium ions with carbonates. After adequate settling, the CaCO3 was separated from the clear liquid using filter paper. This supernatant contained only the chelated ions that cannot be precipitated due to complexation with the chelating agent. A second ICP-OES analysis was carried out on this liquid solution to determine the chelated ions. By comparing the initial total calcium concentration before carbonate separation and after filtration, the difference between the total ions and chelated ions corresponds to the free or unchelated calcium ions. This quantification process is depicted in Fig. 4, where X is the chelated agent (EDTA or GLDA). Equations 2–4 show how carbonates are formed from this process.

$$CaCl_2 \rightarrow Ca^{2+} + Cl^- \quad (3)$$

$$Na_2CO_3 \rightarrow 2Na^+ + CO_3^{2-} \quad (4)$$

$$Ca^{2+} + CO_3^{2-} \rightarrow CaCO_3 \quad (5)$$

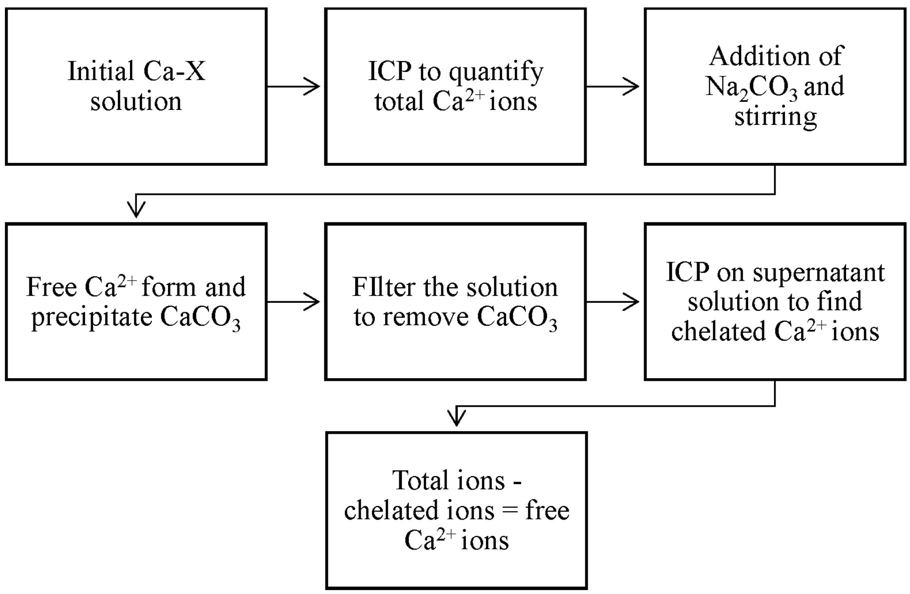

Figure 4—Flowchart of the ICP process to quantify total, free, and chelated calcium ions

Actual case. As given in the preliminary case, a brine of seawater composition was prepared using Table 2. Thereafter, different concentrations of chelating agent were used with brine to test the solubility enhancement of CO_2. Brine and CO_2 were equilibrated in a PVT cell at 70°C and 1500 psi for 24 hours each. After that period, the effluent volume was measured for expansion due to solubilization.

Results and Discussion

Preliminary study

The solubilities of CO_2 in the three separate brine samples were calculated using the ideal gas law and given in Table 4 and Fig. 5. The control sample had a CO_2 solubility of 1.03 mol/kg without any chelating agent. The solubility increased to 1.08 mol/kg with the addition of 5% EDTA, while it was the highest with 1.203 mol/kg for 5% GLDA, indicating the effectiveness of chelating agents in improving the CO_2 uptake.

Table 4—Solubility of CO_2 in brines (preliminary)

Brine sample	**Solubility (mol/kg)**
Control sample (20% C'aC'h)	1.030
20% CaCh+5% EDTA	1.080
20% CaCb+5% GLDA	1.203

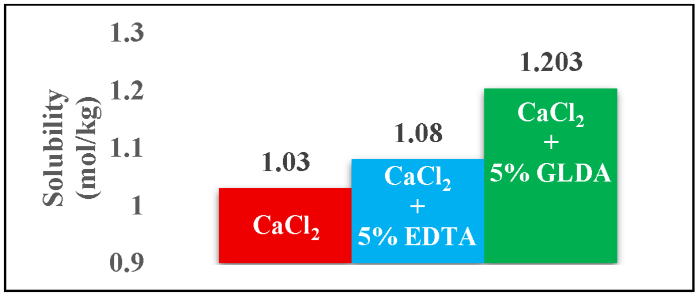

Figure 5—Graphical comparison of CO_2 solubility in different brine samples

Table 5—Ion concentration from ICP

	Ion concentration (mg/L)			
	Total	Chelated	Free	Chelation efficiency
+5% EDTA	929	616	313	66.3%
+5% GLDA	905	860	45	95%

The ICP-OES results for the calcium ions are given in Table 4, showing the chelation effect of both EDTA and GLA on the CaCl2 solution after CO_2 injection. EDTA was able to chelate 66.3% of the calcium ions, while GLDA chelated nearly all calcium ions (95%). This is in line with the solubility results, indicating that the chelating agents improved the CO_2 solubility by chelating the free ions, reducing ionic interactions and preventing salt formation.

Comparing these results with the study done by Mohammed et al. (2025) on the subsurface mineralization of CO_2 in sour brine in the presence of EDTA, some validations can be drawn. At 5% EDTA and 1-10% CaCl2 concentrations from the study, mineralization or precipitation occurred only for 7.5% and 10% CaCl2 concentrations at the same temperature of 50°C and pressure of 500 psi. This implies that increasing CaCl2 composition would result in a higher amount of precipitation for the same quantity of EDTA. However, mineralization only took place when the pH trend decreased. These findings complement the results from the current study, which show the enhancement of CO_2 solubility at similar conditions, followed by the onset of early mineralization that inhibits efficient solubilization. Table 6 shows the CaCl2 concentrations against brine pH from the study, while a comparison Table 7 has been constructed to shed light on these findings and draw conclusions as to what extent the results are relevant at the field level.

Table 6—Subsurface mineralization with 5% EDTA at 50°C (Mohammed et al., 2025)

CaCli %	pH
1	6.2
2	6.4
3	5.9
5	5.8
7.5	5.8
10	5.7

Table 7—Comparison with findings from Mohammed et al. (2025)

Parameters	Study	
	Current	Mohammed et al. (2025)
CaCl2 concentration	20%	1-10%
EDTA concentration	5%	5%
Brine pH	7.59	5.7-6.2
Temperature	50°C	50°C
Pressure	1500 psi	500 psi
Solubility	1.08 mol/kg	Not mentioned
Mineralization	Potential due free cations presence	Yes
Precipitation	Moderate	Yes

For 7.5% and 10% CaCl2 in Table 6, mineralization was possible. The prime contrasting factor here is the pH of brines, which, from the current study (7.59), would fall in the no-mineralization trend. The increased pressure also allows more CO_2 to dissolve, shifting the carbonate speciation equilibrium as cations stay dissolved in brine. It is important to note that the compared study involved H_2S and methane (CH4) as impurities with CO_2, hence it is not appropriate to draw a direct correlation between them.

Study results

The solubility of CO_2 was calculated using Equations 1 and 2 for different chelating agent concentrations and are given in Table 8 and compared with the preliminary results in Fig. 6.

Table 8—Solubility results (present study)

Chelating agent concentration	Solubility (mol/kg)	pH
0%	0.095	7.55
0.1% GLDA	0.263	6.97
0.1% EDTA	0.274	6.05
0.25% GLDA	0.279	5.31
0.25% EDTA	0.264	4.84
0.5% GLDA	0.259	5.36
0.5% EDTA	0.292	4.54
0.75% GLDA	0.476	4.93
0.75% EDTA	0.263	4.43

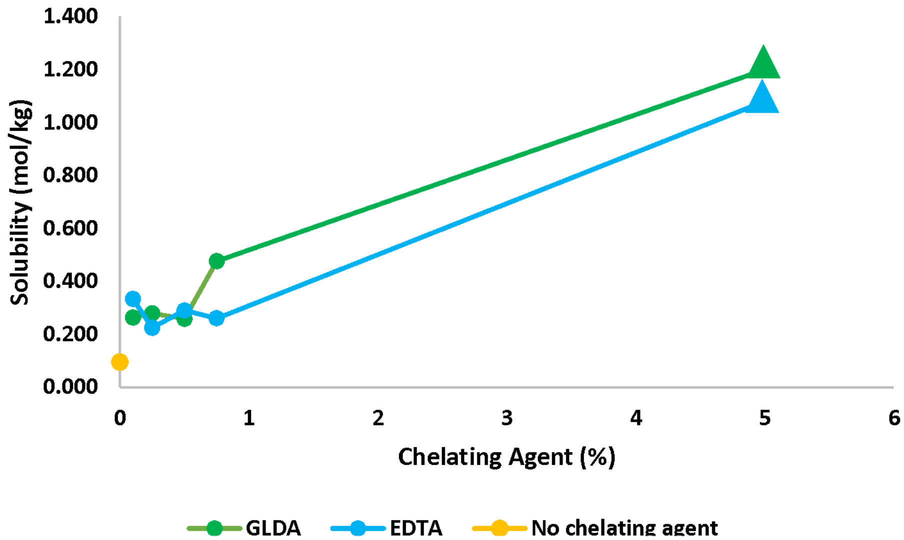

Figure 6—Solubility vs. chelating agent in brine for present (•) and preliminary study (▲)

The decrease in pH also indicates that more carbonic acid is formed due to improved CO_2 retention, and this may also be attributed to the release of H+ ions from the chelating agent itself. On visual inspection, there was precipitation only in the CO_2-brine sample without chelating agents, further confirming that even a small quantity of chelating agents was able to bind a good number of cations. Fig. 7 shows the condition of these mixtures after CO_2 interaction.

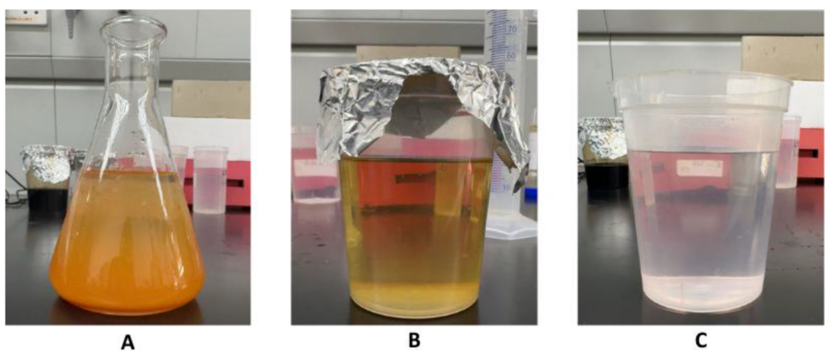

Figure 7—Samples after CO_2 exposure - A) no chelating agent, B) 0.1% GLDA, C) 0.1% EDTA

Conclusions

The use of chelating agents in the carbon sequestration field is not well amplified, and this study presents an approach to improve CO_2 solubility in saline aquifers using these chemicals, especially EDTA and GLDA. The study was able to provide lab-scale results supporting the successful use of chelating agents in binding cations present in brines responsible for salt formation and precipitation, and enhancing CO_2 solubilization by freeing up the space consumed by these cations and reducing ionic activity. In this study, GLDA proved to be a better candidate in terms of improving CO_2 solubility as compared to EDTA. Owing to its biodegradable nature, it becomes a suitable additive for carbon capture and storage. However, carrying out cost-effective operations requires careful selection of the quantity of chelating agent being used due to their expensive nature. Further tests in the study can validate these findings on the effect of chelating agents on mineralization during rock-fluid interactions.

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